

FICTIONAL DEMO SUBMISSION

For LMS grading-workflow demonstration only

Assignment 1 - Function Transformations

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Course	MHF4U - Advanced Functions
Submitted	September 16, 2026
Status	Submitted for grading - demo record

Selected responses

These original responses are intentionally concise and are provided only to demonstrate Moodle submission, download, and grading workflows.

Question 1

For $g(x) = -2(x - 3)^2 + 5$, I move the parent function right 3 and up 5, reflect it across the x-axis, and apply a vertical stretch of 2. The vertex is (3, 5), the axis is $x = 3$, the domain is all real x, and the range is $y \leq 5$.

Question 2

My table for $h(x) = 3|x + 2| - 4$ uses $x = -4, -3, -2, -1, 0$ and gives $y = 2, -1, -4, -1, 2$. The vertex is (-2, -4). The y-intercept is (0, 2), and solving $3|x + 2| = 4$ gives $x = -2 \pm \frac{4}{3}$.

Question 4

Starting with $y = (3x - 7)/5$, I switch x and y: $x = (3y - 7)/5$. Then $5x = 3y - 7$, so $y = (5x + 7)/3$. Substitution gives $f(f^{-1}(x)) = x$.
